Tactix: Rank 12 proof-production recovery

Codex for Andrew

26 September 2026

Outcome. The Rank 12 proof-enabled slowdown is substantially repaired without changing Tactix's native DSR certificate. In balanced, same-node pairs, the current backend took 23.356 and 23.262 seconds; the bounded-spool candidate took 12.188 and 12.338 seconds. Both candidate runs emitted exactly the same 133,019,634-byte proof (SHA-256 `c794f8d973045fecd759d3b64fade0e7e42a9bcc1eeaaa3`) as the previously independent DRAT-trim/CakeLPR-verified current run. The AM337 Rank 12 baseline remains about 11.0 seconds. A full current-source Rank/B28+C5 certificate gate is still blocked by the separate D2F proof failure; this focused performance result does not promote Phase 1.

Cause and repair. Current and AM337 solver work counts are identical on this case, and paired proofless solves are both about 10.6 seconds. An old-Congruence-object ablation left current proof bytes and runtime unchanged. An old-proof-backend ablation restored about 11.1 seconds, locating the cost in proof serialization. A direct counter run measured 177,233 split-unit transactions and exactly 177,233 temporary-file creations, with 9.188 seconds accumulated in the transaction flush routine. The general split-unit transaction is needed to preserve prior D1/H2 native DSR correctness, so the repair keeps its emitted proof events and buffers short deletion blocks in memory. Blocks over 64 KiB still spill to a temporary file. The buffered bytes are appended to the existing binary stream buffer, avoiding a separate write per block.

The first implementation had a 160-byte proof difference: a following RUP unit did not close a deletion block once that block lived in memory rather than a file. That boundary was corrected and added as a byte-exact serializer regression. The final certificate now matches the checked parent byte for byte. Serializer tests also cover the exceptional file-spill path and an external deletion. The public X7 action set, checker options and proof route are unchanged.

Remaining work. Opus AM473 remains the owner of latest-proof-source fuzzing; its terminal public-policy report is still due. The published AM465 D2F internal API reproducer remains a native DSR rejection on current source, and a public control on that CNF follows a different, fully checked path. A general N2 repair must be based on checker-visible proof state and proven public reach, then the full Debug, Opus fuzz, Rank/B28+C5, WV100 and WV2-89 gates must be rerun. N1–N4 and online-learning readiness remain open. The 1,166-instance Tier D run at 5,000 seconds, 4 CPUs and 32 GiB has not been submitted.

Evidence. `benchmarks/reports/2026-09-26-am479-bounded-dsr-spool-rank12.md`; `benchmarks/runs/20260926-am479-bounded-dsr-spool/MANIFEST.md`; `benchmarks/runs/20260926-am475-rank5-12-diagnostic/results/011.json`; `benchmarks/reports/2026-09-26-am476-d2f-current-source-boundary.md`.